

LLMs for NLP

Project Overview

- **Team:** 2 members (recommended) - 4 members maximum.
- **Core Workflow:**
 - 1. **Implement** a model/system addressing a problem.
 - 2. **Evaluate** using appropriate metrics (accuracy, precision, recall, etc.) and compare to state-of-the-art (SOTA) methods.
 - 3. **Document** progress and findings in a formal report and presentation (English, IEEE format).

Comprehensive Milestone: Final Project Delivery

Deadline: 2026-06-08 | **Total Grade:** 9 Points

Objective: Execute an end-to-end research project. This includes establishing the research context, collecting datasets, implementing a baseline, developing an original contribution, and compiling the findings into a well-organized academic report and presentation.

Part 1: Action Items & Research Process

1. Literature Review & State-of-the-Art (SOTA)

- Conduct a thorough literature review (minimum 10 academic papers).
- Formulate at least 1 meaningful research question to address now or in future work.
- Summarize the current landscape, identifying gaps, shortcomings, and potential improvements.

2. Dataset Collection & Baseline Implementation

- Obtain a relevant dataset (e.g., Kaggle, UCI, Papers with Code) and preprocess the data (cleaning, normalization, handling missing values).
- Implement at least one baseline method (e.g., logistic regression, SVM, simple neural network, or pretrained model).
- Evaluate baseline performance using appropriate metrics (accuracy, F1-score, ROC-AUC) to establish a benchmark.

3. Original Contribution

- Develop an original contribution by either:
 - **Addressing a new problem:** Investigating an unexplored medical data science challenge.

- **Improving existing methods:** Optimizing models, applying novel training strategies (e.g., self-supervised learning, data augmentation), proposing a hybrid model, or conducting extensive robustness experiments.
- *Note on Novelty:* Even if your model **doesn't beat the SOTA (most likely)** and even if it **underperforms** previous approaches, following the guidelines (analysis, comparison and grounded conclusions) will lead to **maximum grade**.

Part 2: Deliverables

1. Formal Academic Report (Documentation & Code) Compile your entire project into a polished, 8-page (excluding references) research report in **English**, using the [IEEE format](#). Your code must also be uploaded. The report must include the title, authors, and the following structure:

- **Abstract:** Summary of the project, contributions, and key findings.
- **Introduction:** Problem description, motivation, and background.
- **Related Work:** Summary of SOTA. Must answer: *What datasets/benchmarks are used? What are their limitations? What are the field's shortcomings? How do the papers relate to each other? What architectures/training techniques are noteworthy?*
- **Methodology:** Detail your datasets, preprocessing, algorithms, models, and techniques.
- **Experiments & Own Contribution:** Detail baseline implementation, your original contribution, and how it differs from existing work.
- **Results and Discussion:** Present findings with visualizations (graphs, tables), analyze insights, and compare your contribution against the baseline.
- **Conclusion:** Summarize outcomes, limitations, and future research directions.

2. Final Presentation (6 Minutes) Prepare a visually clear presentation (with slides, figures, graphs, and metrics) covering:

- The topic, problem, context, and research questions.
- Key steps of your methodology.
- Experimental results and your main contributions.
- Conclusions and future research areas.
- *Note:* This presentation will be followed by a Q&A and is used to scale your final project grade based on clarity and depth.

Grading Breakdown (9 Points Total)

1. Literature Review (2.25p)

- **(0.75p) References:** At least 10 academic papers included.
- **(0.50p) Research Questions:** At least 1 meaningful research question provided.
- **(1.00p) Content/Landscape:** Explicitly answers the 5 key sub-questions, worth 0.20p each:
 - What datasets are used and why? (0.2p)
 - What benchmarks or evaluation methods are used? Are there any limitations? (0.2p)
 - What are the current shortcomings in the field? (0.2p)
 - How do the works relate to each other? (0.2p)
 - What architectures and training techniques are used? Any noteworthy or novel approaches? (0.2p)

2. Dataset & Baseline (2.25p)

- **(1.25p) Dataset Description:** Rationale, purpose, record count, quality, collection method, size, and features.
- **(0.25p) Baseline Description:** Clear explanation of the implemented baseline method.
- **(0.50p) Initial Results:** Presentation of baseline performance results.
- **(0.25p) Result Analysis:** Interpretation and insights from the obtained baseline results.

3. Own Contribution (2.25p)

- **(0.75p) Explanation:** Clear, well-justified explanation of your original contribution.
- **(1.00p) Implementation & Results:** Execution on selected datasets.
- **(0.50p) Analysis & Comparison:** Result interpretation and direct comparison with the baseline/SOTA.

4. Final Paper & Presentation (2.25p)

- **(2.25p) Compilation & Delivery:** Adherence to IEEE format, 8-page limit, paper structure, visualization quality, presentation clarity, depth of explanation, and Q&A performance.

Pro-Tips for Success

- **Reading Papers:** Use the 3-pass method to read efficiently. Document main contributions immediately, focus on **contributions** and **limitations**:
<https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf>
- **Sourcing:** Start with recent literature reviews. Explore top conferences (EMNLP, EACL, ACL etc) using the [CORE ranking portal](#) or arXiv. Evaluate papers based on citations, year, and author credibility:
 - https://2025.aclweb.org/program/main_papers/
 - https://2025.emnlp.org/program/main_papers/
 - etc.
- **Engagement:** Choose a topic that genuinely interests you to make the research process enjoyable!

Topics

We recommend choosing an area and direction that most interests you, but be aware of the time limit and hardware limitations. If uncertain, we can help validate them over email.

Some suggestions of research topics that can be implemented without access to extreme hardware resources:

1. Vector quantization for topic modeling
2. Minor sequence-to-sequence learning tasks (e.g. grapheme to phoneme on <http://www.speech.cs.cmu.edu/cgi-bin/cmudict>)
3. Exploring internal representations and architectures for LLMs (e.g. injecting knowledge via small parameter perturbations)
4. Ablation studies for Transformers/GPT/BERT architectures

You can ALSO select (**examples are just informative**, not mandatory):

Active/past shared tasks:

PARSME 2.0 - <https://www.codabench.org/competitions/12002/>

SMM4H-HearD 2026 Task 7 - <https://www.codabench.org/competitions/13991/>

Universal Dependencies Shared Task - <https://universaldependencies.org/conll18/>

Active/past Kaggle competitions